Reliability_Metrics

Metric	Contents	λd (FIT)	λdd (FIT)	MTTFd (y)	DCavg (%)	Target	Margin	Verdict
—	**Reliability metrics — Category 3 verification** Quantitative verification of the selected combination (25119-2, 7.3.5): MTTFd class per Table 4 (medium = 10–30 y per channel), DCavg per Table 5 (medium = 60–90 %), CCF per Annex D (≥ 65). Component failure rates come from the SafeFIT prediction (IEC/TR 62380, mission profile 'Agricultural tractor, cab ECU'); the dangerous fraction and diagnostic credit per failure mode come from the SafeFMEA (7_Analyses). The MTTFd, DCavg and margin columns of this table are live SafeSuite computed attributes — edit a λ value and the derived figures recompute (the norm's conversion 1 FIT = 1e-9/h, year = 8760 h).							—
Channel 1 (L1 + O1 path)	Main MCU, input conditioning ch1, high-side driver, shunt, supply share. λ values aggregated from the SafeFIT project over the FMEA dangerous modes.	245.09	179.41	465.769	73.2017	10	455.769	PASS
Channel 2 (L2 + O2 path)	Monitor MCU, NC-contact path, speed pickup conditioning, low-side MOSFET, watchdog.	159.54	120.39	715.527	75.4607	10	705.527	PASS
Symmetric MTTFdC	**Symmetrised two-channel MTTFd (Annex B.4)** MTTFdC = (2/3) · harmonic combination of the channel values — computed by the analysis stage: 599.5 y (medium). Formula B.4 applied to the two channel MTTFd values above.	190.43		599.46		10	589.46	PASS
CCF score	**CCF score (Annex D)** Separation 15/15 (opposite connector rows, split harness), diversity 20/20 (diverse MCUs + output technologies), design/application 15/20 (overvoltage protection, proven components; over-actuation protection partial), FMEA-informed design 5/5, training 5/5, EMC tested 25/25, other environmental 0/10 (vibration retest pending, OB-3). Total 85 ≥ 65 → PASS.							PASS
FTA residuals	**FTA top-event residuals (mission 8760 h basis)** Top event: Unintended engagement (SG-01) — Residual: 2.9 FIT — Dominated by: CCF residual + undetected driver stage faults Top event: Fails to disengage (SG-02) — Residual: 11 FIT — Dominated by: valve spool jam (hydraulic context) + double stuck-on Top event: Restart without command (SG-03) — Residual: 25.3 FIT — Dominated by: latched-switch + interlock double faults							PASS